VF Asia Footwear Laboratory Testing Manual

Test Method for Cement – Percent Solid	Test method # VF-H04 Page 1 of 1
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PURPOSE AND SCOPE: To measure the % solid of adhesive.

REFERENCE: None

EQUIPMENT AND MATERIAL:

1. Balance, capable of weighing to 0.0001grams.
2. Weighing dish, aluminum or plastics.
3. Air-circulating oven, capable of maintaining 110.0 ± 2.0 ° C.
4. Desiccator, with drying agent.
5. Adhesive or cement in liquid form.

SAMPLE PREPARATION:

Ensure that a well-mixed, homogeneous test material is used. Effective mixing can be performed by manual stirring or suitable mixing equipment.

REPETITIONS: Test two (2) samples per material

TEST PROCEDURE:

NOTE:

- ***Test should be carried out at the standard laboratory condition at 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.***
- ***Solvents used for adhesives or cement are flammable which should be handled carefully. Contact with the skin and inhalation must be avoided. Protective gloves, gas mask and safety glasses should be worn.***

1. Weigh the weighing dish to the nearest 0.0001 grams **(W1).**
2. Pour at least 5 ± 0.1 grams of a test material into the weighing dish.
3. Immediately obtain and record the weight of a test sample and weighing dish to the nearest 0.0001 grams **(W2).**
4. Place the test sample/weighing dish in an oven at 70 ° C for 3 hours.
5. Remove the test sample /weighing dish from the oven. Place in a desiccator for at least 30 minutes, and allow to cool to 23.0 ± 2.0 ° C.
6. Weigh the test sample / weighing dish to the nearest 0.0001 grams **(W3).**

CALCULATION:

% Solid = $(W_3-W_1) / (W_2-W_1) *100$

REPORT: The average of the two readings to the nearest 0.1 percent (%)